

Aircraft Wing Aerodynamic Configuration Variables

Concise reference of wing variables that can be changed to affect aerodynamics. Engine and propeller placement are excluded.

1. Overall wing geometry

Variable	Primary aerodynamic effect
Wing area (S)	Changes lift and wing loading; affects induced/profile drag.
Span (b)	Higher span generally reduces induced drag and improves efficiency.
Aspect ratio (AR)	Higher AR reduces induced drag, especially at low speed.
Mean aerodynamic chord	Affects Reynolds number, lift characteristics and sizing.
Root chord	Changes lift distribution and local Reynolds number.
Tip chord	Changes lift distribution, tip loading and stall behaviour.
Taper ratio	Shapes lift distribution and affects induced drag and stall.
Sweep angle	Reduces compressibility effects at high speed; affects lift and stall.
Quarter-chord sweep	Influences aerodynamic centre and compressibility.
Leading-edge sweep	Strongly affects stall progression and high-speed behaviour.
Trailing-edge sweep	Changes lift distribution and pressure recovery.
Wing thickness	Affects drag, lift characteristics and structural volume.
Thickness distribution	Controls pressure distribution, drag and flow separation.
Camber	More camber generally increases lift and pitching moment.
Camber distribution	Tunes lift, drag and pitching-moment characteristics.
Leading-edge radius	Affects stall onset, peak lift and leading-edge drag.
Trailing-edge angle	Affects pressure recovery and profile drag.

2. Wing positioning and orientation

Variable	Primary aerodynamic effect
Wing incidence angle	Sets wing lift at a given fuselage attitude.
Dihedral angle	Improves lateral stability and roll self-righting.
Anhedral angle	Reduces lateral stability and can increase roll response.
Wing vertical position	Changes wing-body interference and stability characteristics.
Wing longitudinal position	Changes pitching moment, trim and stability.
Wing cant	Changes lift-vector orientation and lateral characteristics.
Wing mounting angle	Changes attitude/trim and wing-fuselage interaction.
Wing-fuselage fairing geometry	Reduces interference drag and flow separation.
Wing-body blending	Can reduce interference drag and improve flow continuity.

3. Wing twist

Variable	Primary aerodynamic effect
Geometric twist	Changes spanwise angle of attack and lift distribution.
Wash-in	Raises tip angle of attack; can increase tip lift but risks tip stall.
Wash-out	Lowers tip angle of attack; delays tip stall and preserves aileron control.
Aerodynamic twist	Changes spanwise aerodynamic loading without simply rotating the wing.
Spanwise incidence variation	Controls spanwise angle of attack and loading.
Spanwise camber variation	Tunes local lift and stall behaviour.
Spanwise airfoil variation	Optimises local performance across the wing.

4. Airfoil configuration

Variable	Primary aerodynamic effect
Airfoil type	Changes lift, drag, stall and pitching-moment characteristics.
Maximum camber	Controls lift capability and pitching moment.
Maximum thickness	Affects drag, structure and flow behaviour.
Thickness-to-chord ratio	Trades structural depth against aerodynamic performance.
Maximum camber location	Changes pressure distribution and pitching moment.
Maximum thickness location	Changes pressure recovery and separation behaviour.
Leading-edge radius	Influences stall and maximum lift.
Trailing-edge thickness	Affects base/profile drag and manufacturability.
Upper-surface curvature	Controls suction and pressure distribution.
Lower-surface curvature	Influences pressure distribution and lift.

Variable	Primary aerodynamic effect
Reflex vs conventional camber	Reflex reduces nose-down pitching moment; conventional camber generally boosts lift.
Supercritical vs conventional airfoil	Supercritical sections improve high-speed compressibility performance.
Laminar-flow airfoil	Can reduce skin-friction drag when laminar flow is maintained.
Spanwise airfoil transitions	Optimise performance for changing local Reynolds/Mach conditions.

5. High-lift devices

Variable	Primary aerodynamic effect
Plain flap	Increases lift and drag through camber change.
Split flap	Produces high lift and high drag.
Slotted flap	Increases maximum lift by delaying separation.
Fowler flap	Increases both camber and effective wing area.
Double-slotted flap	Provides very high lift for take-off/landing.
Triple-slotted flap	Provides extreme high-lift capability.
Flaperon	Combines flap lift increase with roll control.
Drooped aileron	Adds lift while retaining some roll control.
Variable-camber trailing edge	Optimises lift/drag across flight conditions.
Leading-edge slat	Delays leading-edge separation and increases maximum lift.
Krueger flap	Improves high-lift performance by modifying the leading edge.
Leading-edge flap	Increases camber and maximum lift.
Drooped leading edge	Increases leading-edge lift and delays stall.
Flap chord	Larger chord generally increases high-lift authority and drag.
Flap span	Controls how much of the wing receives high-lift augmentation.
Flap deflection	Higher deflection usually increases lift and drag until separation limits it.
Flap position	Changes effectiveness and local flow.
Flap gap	Controls slot flow and separation delay.
Flap overlap	Changes slot geometry and high-lift effectiveness.
Number of slots	More slots can increase maximum lift but add complexity and drag.
Leading-edge device chord	Larger chord generally increases high-lift authority.
Leading-edge device span	Controls spanwise extent of leading-edge lift enhancement.
Leading-edge device deflection	Changes lift, drag and stall margin.
Leading-edge device gap	Controls energising flow over the upper surface.
Leading-edge device position	Changes pressure distribution and separation behaviour.

6. Control surfaces

Variable	Primary aerodynamic effect
Aileron area	Changes roll control authority.
Aileron chord	Changes roll authority and hinge moment.
Aileron span	Changes total roll control authority.
Aileron deflection	Changes rolling moment and drag.
Aileron hinge position	Changes control effectiveness and hinge moment.
Elevon area	Controls combined pitch and roll authority.
Elevon chord	Changes pitch/roll authority and drag.
Elevon span	Changes control authority.
Elevon deflection	Changes pitch and roll moments.
Spoiler area	Increases drag and can reduce lift.
Spoiler deflection	Controls lift dump and drag.
Spoileron deflection	Produces asymmetric drag/lift for roll control.
Airbrake area	Increases drag for speed control.
Airbrake deflection	Controls drag increase.
Control-surface location	Changes moment arm and effectiveness.

7. Wingtip configuration

Variable	Primary aerodynamic effect
Tip shape	Changes tip loading, vortex formation and induced drag.
Rounded tip	Can smooth loading and reduce local separation.
Square tip	Simple but can increase tip losses.

Variable	Primary aerodynamic effect
Hoerner tip	Can reduce tip vortex losses with simple geometry.
Raked tip	Increases effective span and can reduce induced drag.
Blended wingtip	Smoothly modifies loading and can reduce drag.
Winglet	Reduces effective induced drag by altering tip vortex flow.
Canted winglet	Tunes induced-drag reduction and structural loads.
Split winglet	Targets vortex reduction with potentially lower height.
Sharklet	Winglet form designed to improve cruise efficiency.
Tip fence	Controls crossflow and tip vortex development.
Endplate	Reduces pressure leakage around the tip.
Winglet height	Higher can improve vortex control but increases weight/load.
Winglet cant	Changes aerodynamic efficiency and loads.
Winglet sweep	Tunes high-speed performance and vortex interaction.
Winglet toe angle	Changes local loading and vortex behaviour.
Winglet twist	Optimises winglet loading across operating conditions.
Winglet taper	Changes loading, drag and structural efficiency.

8. Variable-geometry and morphing wings

Variable	Primary aerodynamic effect
Variable sweep	Trades low-speed efficiency against high-speed performance.
Variable span	Can trade induced drag against structural/operational constraints.
Folding wings	Primarily changes operational footprint; can also alter effective span.
Telescoping wings	Changes span/aspect ratio and induced drag.
Variable chord	Changes wing area, loading and aerodynamic characteristics.
Morphing wing	Continuously changes geometry to optimise performance.
Variable camber	Adapts lift/drag performance to flight condition.
Variable twist	Optimises spanwise loading across conditions.
Variable wing incidence	Changes trim and lift without changing the basic wing.
Adaptive airfoil	Changes local section shape to optimise performance.
Flexible/compliant wing	Allows passive or active shape adaptation and load alleviation.

9. Surface and flow-control features

Variable	Primary aerodynamic effect
Surface roughness	Usually increases skin-friction drag and can trigger earlier transition.
Boundary-layer trip	Forces transition from laminar to turbulent flow.
Vortex generators	Energise boundary layer and delay separation, with added drag.
Wing fences	Restrict spanwise flow and can delay stall.
Ridges	Can alter boundary-layer behaviour; effect depends strongly on geometry.
Riblets	Can reduce turbulent skin-friction drag when correctly sized/aligned.
Leading-edge serrations	Modify vortical flow and can delay/reshape separation.
Leading-edge cuffs	Improve stall characteristics and maximum lift.
Stall strips	Force predictable/localised stall onset.
Boundary-layer suction	Removes low-energy flow and can delay separation.
Boundary-layer blowing	Adds momentum to the boundary layer and can delay separation.